Roadmap to Large-Scale Modular Quantum Computing: Base, Aggressive, and Stretch Scenarios Through 2030

Technical Research Paper v2.1

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Abstract

This paper presents a gated development roadmap for QONTOS through 2030. It replaces a single-path million-qubit storyline with three scenarios: `Base`, `Aggressive`, and `Stretch`. The stretch scenario retains the long-term aim of **1,000,000 physical qubits** and **10,000 logical qubits**, but the paper now treats that outcome as conditional on explicit subsystem breakthroughs in device performance, error correction, interconnects, cryogenics, and software control. The purpose of the roadmap is therefore not to guarantee a result. It is to define a disciplined sequence of gates, fallback outcomes, and decision points.

Claim status: program roadmap and scenario planning document

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1. Roadmap Philosophy

The previous roadmap framing was too singular. A technically feasible program needs:

- base outcome
- aggressive outcome
- stretch outcome
- validation gates
- no-go conditions
- fallback paths

1.1 Scenario Summary

Scenario	By 2030 likely outcome
Base	strong software, digital twin, modular hardware integration, early logical qubits
Aggressive	large modular machine with meaningful logical-qubit and distributed FTQC demonstrations
Stretch	1,000,000 physical qubits and 10,000 logical qubits

1.2 What This Roadmap Does Not Claim

This roadmap does not claim:

- that the stretch scenario is already validated
- that the stretch budget is guaranteed to be sufficient
- that all subsystem breakthroughs will land on the same timeline

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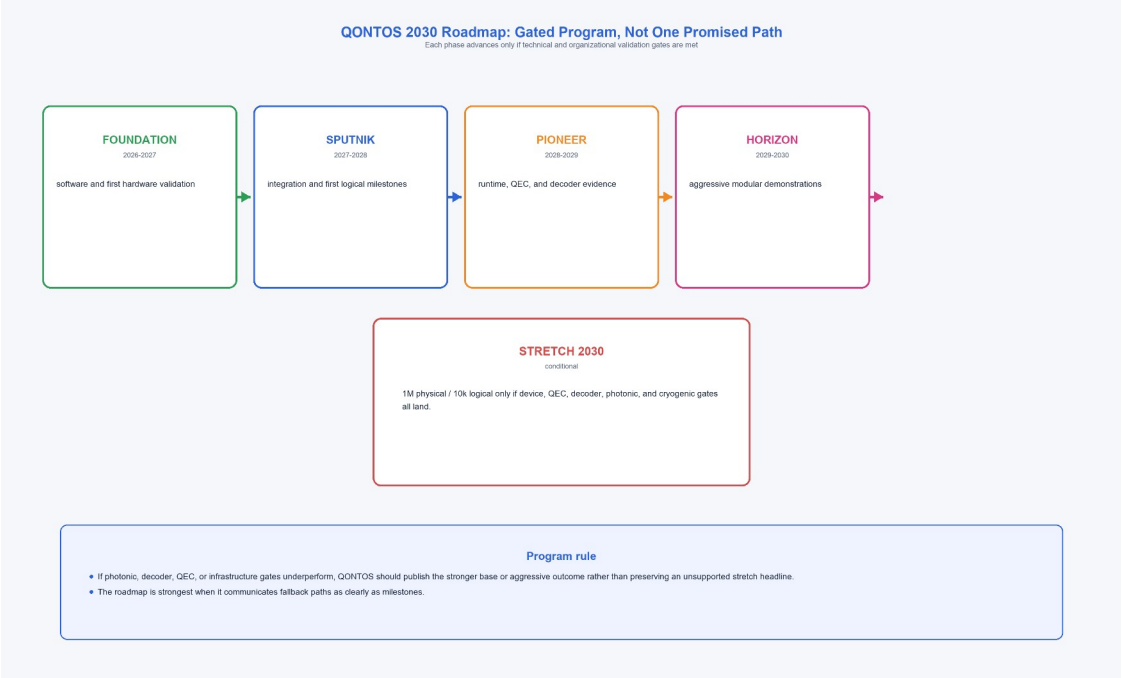
2. Five Program Phases



Gated Roadmap

Figure 1: The roadmap should be read as a gated program with fallback outcomes, not as one promised path.

2.1 FOUNDATION (2025-2026)

Primary objective

- establish the software, benchmark, device, and digital-twin foundation

Base outcome

- strong software-first orchestration platform
- current benchmark harness and replayability
- improved architecture and assumptions discipline

Aggressive outcome

- first hardware-linked validation path

- stronger device and packaging evidence

Stretch relevance

- first evidence that stretch device and QEC assumptions are not obviously impossible

Validation gates

- canonical architecture constants fixed
- benchmark methodology paper in place
- first device assumptions table grounded in literature
- current platform and simulation artifacts reproducible

No-go conditions

- no coherent benchmark program
- no plausible device story
- no current platform credibility

Fallback if gate fails

- remain a software-first orchestration and digital-twin platform while hardware path matures

2.2 SPUTNIK (2026-2027)

Primary objective

- move from architecture thesis to first scaled modular hardware assumptions

Base outcome

- small modular hardware integration path
- digital-twin and software stack mature together

Aggressive outcome

- 10,000 physical-qubit class module assumptions become defensible
- first 100 logical-qubit program target becomes credible

Stretch relevance

- first evidence toward the 10,000 physical-qubit module target

Validation gates

- packaging and thermal budgets reviewed
- first logical-qubit milestone path defined
- inter-module entanglement experiment plan credible

No-go conditions

- module thermal/control assumptions fail
- device packaging does not preserve performance

Fallback if gate fails

- reduce module size and extend timeline while preserving modular software roadmap

2.3 PIONEER (2027-2028)

Primary objective

- validate multi-module behavior and distributed execution

Base outcome

- mature distributed runtime and modular benchmark program

Aggressive outcome

- large multi-module demonstrations
- 100-1,000 logical-qubit path becomes plausible

Stretch relevance

- first credible path to 100,000 physical qubits and modular FTQC subroutines

Validation gates

- distributed runtime and digital twin aligned
- inter-module communication passes minimum thresholds
- decoder path supports the targeted correction-cycle budget

No-go conditions

- communication overhead dominates
- decoder latency or reliability collapses at modular scale

Fallback if gate fails

- focus on lower-scale modular advantage and software commercialization

2.4 HORIZON (2028-2029)

Primary objective

- translate modular maturity into large-scale logical performance

Base outcome

- strong modular system platform with useful workload demonstrations

Aggressive outcome

- 100,000 physical-qubit and 100-1,000 logical-qubit class system becomes feasible

Stretch relevance

- 500,000 physical qubits and 5,000 logical qubits become conditionally plausible

Validation gates

- effective overhead moves materially below conservative baselines
- photonic and cryogenic assumptions survive systems validation
- benchmark ladder shows meaningful upward progression

No-go conditions

- effective overhead remains near 1000:1
- photonic path remains too weak for modular scaling

Fallback if gate fails

- keep aggressive system target, retire or defer the stretch case

2.5 SUMMIT (2029-2030)

Primary objective

- determine whether the stretch architecture is truly reachable

Base outcome

- commercially meaningful modular platform with validated logical resources

Aggressive outcome

- large fault-tolerant modular machine with strong flagship benchmark progress

Stretch outcome

- 1,000,000 physical qubits
- 10,000 logical qubits
- flagship chemistry or comparable application advantage

Validation gates

- device assumptions support stretch performance
- effective overhead near stretch case
- interconnect assumptions near stretch case
- benchmark and replay evidence support the claim set

No-go conditions

- one or more critical subsystems remain at base or conservative performance

Fallback if gate fails

- ship the strongest aggressive architecture achievable and continue stretch R&D beyond 2030

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3. Scenario Tables

3.1 Architecture Outcome by 2030

Metric
Physical qubits
Logical qubits
Effective overhead
Interconnect regime

3.2 Investment Envelope

Scenario	Indicative interpretation
Base	software-led and hardware-integrating program
Aggressive	full modular research and scale-up program
Stretch	approximately the previously cited $530-780M class envelope, contingent on subsystem success

The stretch investment number should remain an estimated planning envelope, not a guarantee of sufficiency.

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4. Risks and Dependencies

4.1 Major technical dependencies

- millisecond-class device regime
- low-enough effective overhead
- practical decoder latency
- adequate transduction and Bell-pair performance
- cryogenic and control scalability

4.2 Major program dependencies

- fabrication partners
- packaging and supply chain
- sustained capital access
- team build-out across software, hardware, cryogenics, and QEC

4.3 Major communication risk

The biggest communication risk is presenting the stretch scenario as if it were already a technically consolidated production plan. This paper should explicitly avoid that.

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5. Recommended External Framing

The most technically credible public statement is:

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QONTOS is pursuing a modular quantum roadmap with base, aggressive,
and stretch scenarios through 2030. The stretch scenario targets 1
million physical qubits and 10,000 logical qubits, contingent on major
advances in qubits, error correction, interconnects, cryogenics, and
control systems.
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Avoid:

- `QONTOS will have 1 million qubits by 2030`
- `QONTOS will solve FeMoco by 2030`
- `The quantum future is 5 years away`

unless explicitly framed as stretch-case language.

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6. Conclusion

This roadmap is strongest when it is used as a disciplined program map rather than a single-path promise. The stretch scenario remains valuable because it defines a bold north star. But the base and aggressive scenarios are equally important because they preserve technical credibility and create real decision points.

The roadmap is therefore feasible only if it remains gated, scenario-based, and tightly linked to benchmark evidence.

References

[1] Internal QONTOS architecture constants, assumptions register, and claim-status guidance.

[2] Representative literature on modular quantum systems, error correction, and large-scale superconducting roadmaps.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Scenario-based roadmap with explicit gates and fallback paths